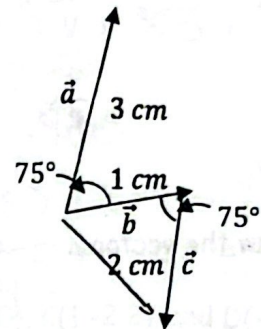


MCV 4U1- Unit 6 Test PM - Introduction to Vectors

Name _____

Date May 12, 2025

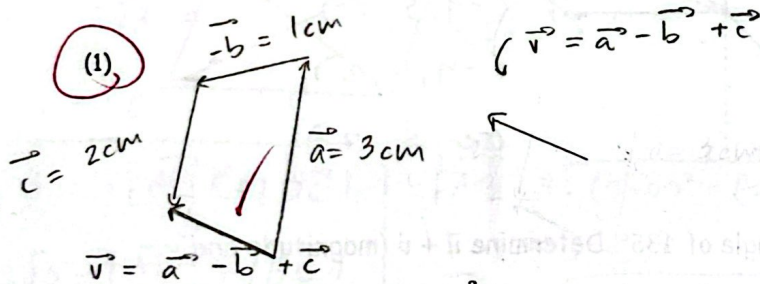
For questions 1-3, Use the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ shown on the right.



1. Determine $|\vec{b} + \vec{c}|$

$$|\vec{b} + \vec{c}| = \sqrt{1^2 + 2^2 - 2(1)(2)\cos 75} = 1.99$$

2. Draw $\vec{v}$ if $\vec{v} = \vec{a} - \vec{b} + \vec{c}$



3. True or False: $\vec{a} = -\frac{3}{2}\vec{c}$. State whether the statement is true. Explain your answer

TRUE. $\vec{a}$ and $\vec{c}$ are opposite vectors so by adding a (-) it makes them have the same direction. The magnitude of $\vec{c}$ multiplied by $\frac{3}{2}$ (bc $\vec{a} = 3$ and $\vec{c} = 2$) makes the statement $\vec{a} = -\frac{3}{2}\vec{c}$ TRUE

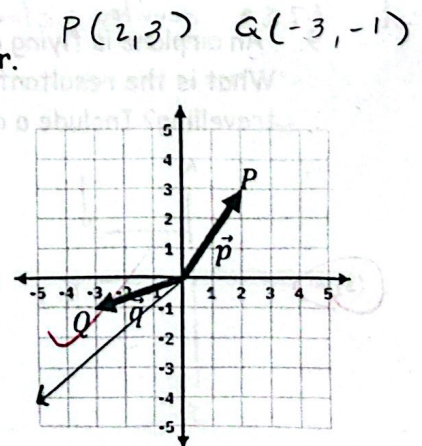
For questions 4-5, use the vectors $\vec{p}$ and $\vec{q}$ shown on the right.

4. Write $\overrightarrow{PQ}$ in component form, and draw the associated position vector.

$$\begin{aligned}\overrightarrow{PQ} &= \vec{Q} - \vec{P} \\ &= (-3 - 2, -1 - 3) \\ &= (-5, -4)\end{aligned}$$

5. State $\vec{p} + \vec{q}$ using standard unit vectors.

$$\begin{aligned}\vec{p} + \vec{q} &= (2\hat{i} + 3\hat{j}) + (-3\hat{i} - \hat{j}) \\ &= (-\hat{i} + 2\hat{j})\end{aligned}$$



6. Using the square based prism shown, simplify $\overrightarrow{PQ} + \overrightarrow{QR} - \overrightarrow{TV} + \overrightarrow{PW}$ as much as possible. Draw a diagram to verify.

(2) $\overrightarrow{PQ} + \overrightarrow{QR} - \overrightarrow{TV} + \overrightarrow{PW}$

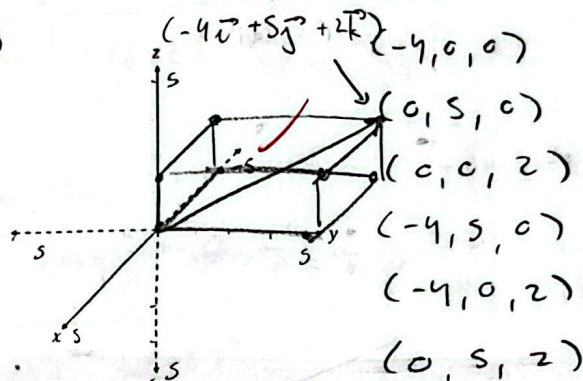
$\overrightarrow{PQ} + \overrightarrow{QR} + \overrightarrow{VT} + \overrightarrow{PW}$

$\overrightarrow{PW}$
 $\overrightarrow{QW}$

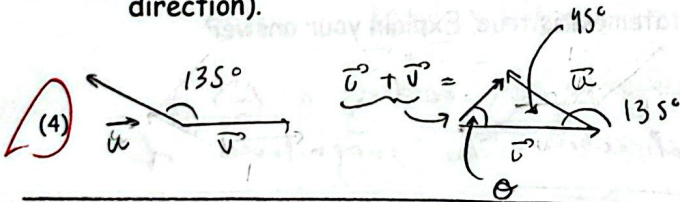
$\overrightarrow{RP}$

7. Draw the vector $\vec{A} = -4\vec{i} + 5\vec{j} + 2\vec{k}$ using the rectangular prism method.

$\vec{A} = (-4, 5, 2)$



8. $\vec{u}$ and $\vec{v}$ are both unit vectors that make an angle of 135° . Determine $\vec{u} + \vec{v}$ (magnitude and direction).



$|\vec{u} + \vec{v}| = \sqrt{1^2 + 1^2 - 2(1)(1)\cos 45^\circ}$
 $= 0.765$

$\frac{\sin 45^\circ}{0.765} = \frac{\sin \theta}{1}$

$\theta = \sin^{-1}\left(\frac{\sin 45^\circ}{0.765}\right) = 67.5^\circ$

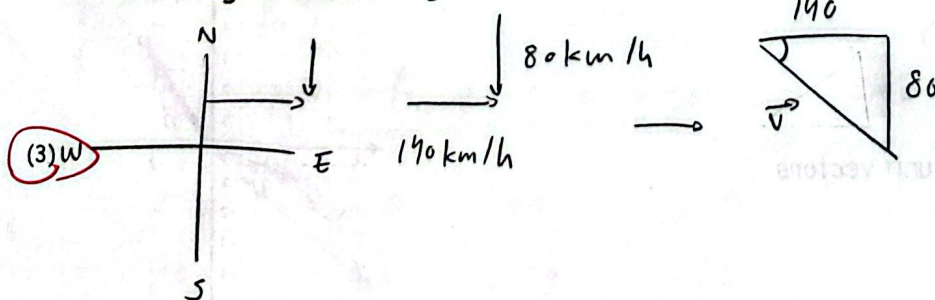
$|\vec{u} + \vec{v}| = 0.765$

direction of $|\vec{u} + \vec{v}|$ is from $\vec{v}$

67.5° counter clockwise

9. An airplane is flying due east at 140 km/h when it encounters a wind of 80 km/h from the north.

What is the resultant ground velocity of the airplane and the direction that the plane is travelling? Include a diagram



$\vec{v} = \sqrt{80^2 + 140^2}$
 $= 20\sqrt{65}$

$\theta = \tan^{-1}\left(\frac{80}{140}\right)$

$= [E 30^\circ S]$

the resultant $\vec{v} = 20\sqrt{65} \text{ km/h} = 161.24 \text{ km/h}$

the direction is $[E 30^\circ S]$

10. If $a(0, -1, -3) + b(1, 1, 1) - c(1, 2, 5) = (-2, 3, -8)$, determine a, b and c

Crystal be

$$a(0, -1, -3) + b(1, 1, 1) - c(1, 2, 5) = (-2, 3, -8)$$

$$(4) (0, -a, -3a) + (b, b, b) + (-c, -2c, -5c) = (-2, 3, -8)$$

$$\begin{matrix} i & j & k \\ 0 + b - c = -2 & -a + b - 2c = 3 & -3a + b - 5c = -8 \end{matrix} \quad (1) \quad (2) \quad (3)$$

$$(1) \quad b = -2 + c \quad (1) \rightarrow (2) \quad -a + (-2 + c) - 2c = 3 \quad (1) + (2) \rightarrow (4)$$

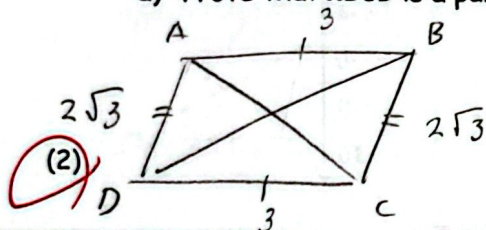
$$-a - 2 + c - 2c = 3 \quad -3(-c - 5) + (-2 + c) - 5c = -8$$

$$3c + 15 - 2 + c - 5c = -8$$

$$-c = -21$$

11. The vertices of a quadrilateral ABCD are given as $A(0, -4, 0)$, $B(3, -4, 0)$, $C(1, -2, 2)$, and $D(-2, -2, 2)$

a) Prove that ABCD is a parallelogram.



$$3 = |\vec{AB}| = |\vec{DC}|$$

$$2\sqrt{3} = |\vec{AD}| = |\vec{BC}|$$

$$|\vec{AB}| = \sqrt{(3-0)^2 + (-4+4)^2 + (0-0)^2}$$

$$= 3$$

$$|\vec{DC}| = \sqrt{(1+2)^2 + (-2+2)^2 + (2-2)^2} = 3$$

$$|\vec{AD}| = \sqrt{(-2-0)^2 + (-2+4)^2 + (2-0)^2} = 2\sqrt{3}$$

b) State an equation of the plane that contains the points A and B.

$$y = -4$$

$$z = 0$$

$$|\vec{BC}| = \sqrt{(1-3)^2 + (-2+4)^2 + (2-0)^2} = 2\sqrt{3}$$

c) State a unit vector in the same direction as $\vec{AB}$

$$\vec{AB} = (3-0, -4-(-4), 0-0)$$

$$= (3, 0, 0)$$

$$|\vec{AB}| = \sqrt{3^2}$$

$$= 3$$

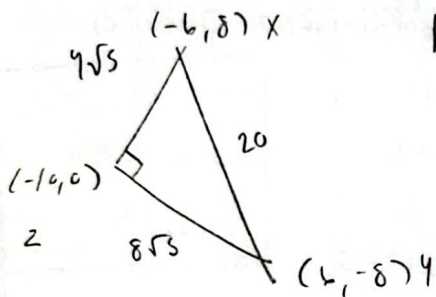
$$\text{unit vector} : \frac{1}{3} (3, 0, 0)$$

12. The three points $X(-6,8)$, $Y(6,-8)$ and $Z(-10,0)$ are on a circle with radius 10 and centre at the origin. Points X and Y are endpoints of a diameter and point Z is on the circle. Using your understanding of vectors, show that X, Y , and Z are the vertices of a right triangle. Include a diagram in your solution.

to be vertices of a right triangle:

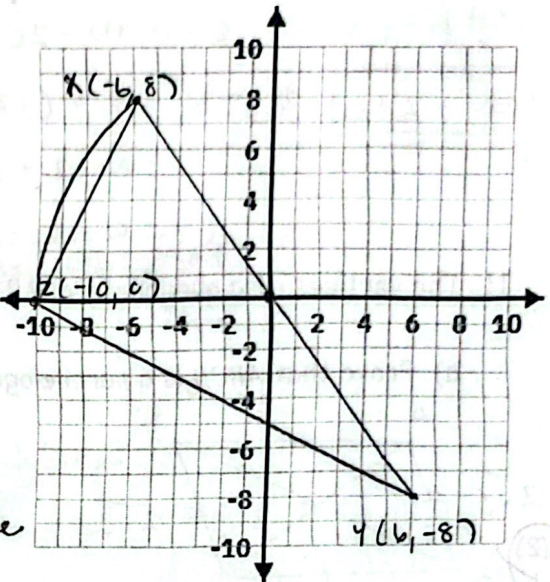
(4) $\vec{XY}$ must be the side with the greatest magnitude

$$\angle Z = 90^\circ$$



$$|\vec{XY}| = \sqrt{(6+6)^2 + (-8-8)^2} = 20$$

double the radius, longest side



cosine law

sin law

$$|\vec{XZ}| = \sqrt{(-10+6)^2 + (0-8)^2} = 4\sqrt{5} \Rightarrow 8.94$$

$$|\vec{ZY}| = \sqrt{(6+10)^2 + (-8-0)^2} = 8\sqrt{5} \Rightarrow 17.88$$

$$20^2 = (8\sqrt{5})^2 + (4\sqrt{5})^2 - 2(8\sqrt{5})(4\sqrt{5})\cos\theta$$

$$\theta = \cos^{-1} \left(\frac{20^2 - (8\sqrt{5})^2 - (4\sqrt{5})^2}{-2(8\sqrt{5})(4\sqrt{5})} \right) = 90^\circ$$

the angle at point $Z = 90^\circ \Rightarrow$ right angle triangle

$$\frac{\sin 90}{20} = \frac{\sin \theta}{4\sqrt{5}}$$

$$\frac{\sin 90}{20} = \frac{\sin \theta}{8\sqrt{5}}$$

$$\theta = \sin^{-1} \left(\frac{4\sqrt{5}(\sin 90)}{20} \right)$$

$$\theta = \sin^{-1} \left(\frac{8\sqrt{5}(\sin 90)}{20} \right)$$

$$\angle Y = 26.56$$

$$\angle X = 63.43$$

$$\begin{aligned} \text{all interior angles} \\ &= 90 + 26.56 + 63.43 \\ &= 180^\circ \end{aligned}$$

all interior angles add up to 180°